WAiK Product Overview

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Executive Summary

WAiK (pronounced "wake") is an AI-powered incident reporting and investigation platform purpose-built for senior care facilities. It transforms the traditionally burdensome process of documenting falls and other incidents from a 45-60 minute paperwork ordeal into a streamlined, voice-enabled experience that takes under 5 minutes—while producing reports that are more complete, more compliant, and more actionable than traditional methods.

> **"The nurse's burden is 5 minutes, but the system does 2 hours of expert-level investigative work for them in the background."**

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The Problem

The Incident Reporting Crisis in Senior Care

Every day, nurses and caregivers in senior living facilities face an impossible choice: spend time caring for residents or spend time documenting incidents. Currently, documentation loses—and so does compliance.

The Numbers

Metric	Reality
Time per incident report	45-60 minutes
Typical compliance field completion	40-60%
Staff complaint #1	Documentation burden
Delayed reports	Hours to days after incident

Metric	Reality
Memory decay impact	Critical details forgotten

What Happens Today

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INCIDENT OCCURS
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      ▼
┌──────────────────────────────────────────────────────────────────┐
│                    STAFF MEMBER AT SCENE                         │
│                                                                  │
│   Must handle BOTH:                                              │
│   ├── 🏥 Resident care (priority)                                │
│   └── 📝 Complete documentation (45-60 minutes)                  │
│                                                                  │
│   What Actually Happens:                                         │
│   ├── ⏰ Staff prioritizes resident care                         │
│   ├── 📋 Report gets pushed to "later"                           │
│   ├── 🧠 Memory fades, details lost                              │
│   ├── 📝 Generic forms miss context-specific questions           │
│   └── 😓 Staff burnout from documentation burden                 │
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INCOMPLETE REPORT → Compliance Issues → Regulatory Fines → Repeat
Incidents
```

Key Pain Points

1. **Time Burden**: 45-60 minutes diverted from patient care per incident
2. **Generic Forms**: Same questions for wheelchair falls and bed falls—missing critical context-specific information
3. **Memory Decay**: Critical details forgotten hours or days later
4. **Staff Burnout**: Documentation is the #1 complaint in nursing
5. **Incomplete Data**: 40-60% of compliance fields left empty
6. **Delayed Reporting**: Reports filed hours or days after the incident
7. **Compliance Risk**: Incomplete reports lead to regulatory issues and liability exposure

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The WAiK Solution

The Two-Agent Architecture

WAiK solves the documentation crisis through a revolutionary **Two-Agent Agentic System** that separates what humans must do from what AI can do better.

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│
│    │                    AGENT 2: "THE INVESTIGATOR"
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│    │                    ──────────────────────────────
│    │
│    │    🔍  Asynchronous, background
│    │
│    │    🧠  Expert-level analysis
│    │
│    │    📊  Compliance-driven
│    │
│    │
│    │    Automatically:
│    │
│    │    ├── Classifies incident subtype (wheelchair, bed, slip, lift)
│    │
│    │    ├── Identifies gaps in the narrative
│    │
│    │    ├── Generates expert follow-up questions
│    │
│    │    └── Queues questions for staff to answer later
│    │
│    │
│    │    All done in ~15 seconds, no human input needed
│    │
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│
│    │                      STAFF ANSWERS LATER
│    │
│    │                      ───────────────────
│    │
│    │    ☕ On break or when convenient                                      │
│
│    │    📝  Via dashboard with targeted questions
│    │
│    │    🎤  Text or voice input
│    │
│    │
│    │    ~5-10 additional minutes (non-urgent, asynchronous)
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│    │                       COMPLETE REPORT
│    │                       ───────────────
│    │
│    │    ✅ 90%+ compliance fields filled
│    │
│    │    ✅ Expert-level detail
│    │
│    │    ✅ Staff time: ~10-15 minutes total (vs 45-60 traditional)
│    │
│    │    ✅ AI-generated insights available
│    │
│    │    ✅ Audit-ready documentation
│    │
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The Result: Better Reports in Less Time

Metric	Traditional	WAiK
Time at scene	45-60 minutes	5 minutes
Total staff time	45-60 minutes	~10-15 minutes
Compliance rate	40-60%	90%+
Question quality	Generic checklist	Expert, context-specific
Subtype handling	Same form for all	Specialized per incident type
Gap identification	Manual review	Automatic AI analysis
Staff experience	Burdensome paperwork	Natural conversation

How It Works

Step 1: Voice Report (5 Minutes)

Staff speaks naturally about what happened. The Report Agent:

1. **Captures the narrative** via voice transcription
2. **Enhances the narrative** using AI to create professional clinical documentation
3. **Creates the incident record** immediately
4. **Notifies administrators** in real-time

Example input (spoken):

> "I found Mrs. Chen on the floor next to her wheelchair. She says her hip hurts. The brakes weren't locked on the chair. She's awake and talking but seems upset."

AI-enhanced output:

> On December 14, 2024, staff member Sarah Johnson discovered resident Margaret Chen on the floor adjacent to her wheelchair in Room 204A. The resident was alert and oriented upon discovery. Key observations: Resident complaining of right hip discomfort; wheelchair brakes were not engaged at time of discovery; resident appeared emotionally distressed but cognitively intact.

Step 2: Automatic Investigation (15 Seconds)

The Investigation Agent immediately:

1. **Classifies the incident subtype**: `fall-wheelchair`
2. **Loads expert question templates** for wheelchair falls
3. **Generates tailored follow-up questions** specific to this incident
4. **Queues questions** for staff to answer on their break

Generated questions (wheelchair fall specific):

- Were the wheelchair brakes engaged prior to the transfer attempt?
- Was the resident using any safety belt or lap buddy when seated?
- Has the wheelchair been inspected for maintenance issues recently?
- Was the cushion positioned correctly before the incident?
- Was the footrest raised or removed before the resident tried to stand?
- Has physical therapy evaluated the resident's wheelchair fit in the last 30 days?

Step 3: Gap Filling (When Convenient)

Staff answers the queued questions from their dashboard—via text or voice—whenever they have time. The Expert Investigator:

1. **Scores the report** in real-time (0-100)
2. **Identifies remaining gaps** based on compliance requirements
3. **Generates new questions** dynamically based on answers
4. **Finalizes** when all required fields are documented

Step 4: AI-Powered Reports & Intelligence

Administrators get:

- **AI-generated reports**: Summary, insights, recommendations, action items
- **Compliance scoring**: Real-time visibility into documentation completeness
- **Intelligence Q&A**: Ask questions about any incident using semantic search (RAG)
- **Pattern analysis**: Cross-incident insights (roadmap)

Core Value Propositions

For Staff

Benefit	Impact
75% less time	~10 minutes vs 45-60 minutes per incident
Natural voice input	Talk naturally, no forms to fill
Asynchronous answers	Respond when convenient, not under pressure
Guided questions	Know exactly what information is needed

For Administrators

Benefit	Impact
Complete reports	90%+ compliance field completion
Expert questions	Context-specific, not generic
Real-time visibility	Incidents appear immediately on dashboard
AI insights	Automated analysis and recommendations
Audit readiness	Documentation meets regulatory standards

For Facilities

Benefit	Impact
Reduced compliance risk	Complete documentation = fewer regulatory issues
Lower liability exposure	Thorough incident documentation protects legally
Improved staff retention	Documentation burden is #1 complaint in nursing
Better root cause analysis	Complete data enables prevention
Cost savings	Staff time redirected to patient care

Technological Moat

1. Two-Agent Agentic Architecture

WAiK's core innovation is the **separation of time-critical capture from thorough analysis**.

Why it matters:

- No other system distinguishes between "what must happen now" and "what can happen later"
- This architecture fundamentally changes the user experience from burden to conversation
- The handoff pattern ensures no data is lost even if the investigation fails

Implementation details:

- Report Agent: LangGraph-style node-based workflow with streaming events
- Investigation Agent: Automatic classification + template-based + AI-tailored question generation
- Expert Investigator: Conversational, session-based, real-time scoring

2. Gold Standards System

WAiK uses **expert-defined compliance checklists** (Gold Standards) that define exactly what information should be captured for each incident type.

Why it matters:

- Not just a generic form—subtype-specific requirements (wheelchair vs bed vs slip vs lift)
- Enables automatic gap detection: AI compares narratives against standards
- Drives intelligent question generation: questions target missing fields
- Powers compliance scoring: real-time visibility into documentation completeness

Implementation details:

- 22+ global fall fields (required for all falls)
- 4-11 additional subtype-specific fields per incident type
- Weighted scoring algorithm (critical fields weighted 2x)
- Extensible: new subtypes can be added

3. Intelligent Question Generation

WAiK doesn't ask generic questions—it generates **expert-level, context-specific questions**.

Why it matters:

- A wheelchair fall needs different questions than a bed fall
- Questions are tailored to the specific narrative (mentions brakes → asks about brake status)
- Deduplication prevents asking the same question twice
- Questions are assigned to the original reporter for continuity

Implementation details:

- Expert question templates per subtype (8 templates each)
- AI tailoring: LLM customizes templates to match the narrative
- Deduplication: checks against existing questions before queuing
- Priority weighting: most important questions asked first

4. Voice-First Experience

WAiK is designed for **voice input at every stage**.

Why it matters:

- Staff can report while attending to residents
- Natural conversation vs structured form filling
- Hands-free operation for high-pressure situations
- Reduces friction to near-zero

Implementation details:

- Web Speech API for speech-to-text (Chrome/Edge)
- Speech Synthesis API for voice prompts
- Companion mode: full hands-free conversational reporting
- Voice answer mode: answer questions via speech

5. RAG-Powered Intelligence

WAiK implements **Retrieval-Augmented Generation (RAG)** for semantic search across incident data.

Why it matters:

- Ask natural language questions about any incident
- Finds relevant information even with imprecise queries
- Enables cross-incident pattern discovery
- Powers the Intelligence Q&A feature

Implementation details:

- text-embedding-3-small for 1536-dimension vectors
- Cosine similarity for semantic search
- Embedding cache for performance
- Question + answer pairs vectorized for comprehensive search

6. Real-Time Compliance Scoring

WAiK calculates **compliance scores in real-time** as reports are completed.

Why it matters:

- Administrators see documentation completeness at a glance
- Staff know when they're "done" (score reaches threshold)
- Gaps are visible before audits, not during them
- Enables proactive improvement

Implementation details:

- Two scoring algorithms: simple completeness + weighted score
- Critical fields weighted 2x (injuries, physician notification)
- Subtype-specific fields weighted 1.5x
- Human-readable gap labels for actionable feedback

Process-Based Moat

1. Expert Knowledge Embedded in Software

WAiK encodes **risk management expertise** directly into the system.

What we've captured:

- Expert-defined compliance requirements (Gold Standards)
- Subtype classification logic (wheelchair, bed, slip, lift)
- Question templates based on what investigators actually ask
- Scoring algorithms based on regulatory priorities

Why it's a moat:

- Competitors would need to recreate this domain expertise
- Our Gold Standards are continuously refined based on real-world usage
- The classification and question logic represents years of risk management knowledge

2. The Asynchronous Workflow Pattern

WAiK pioneered the **"capture now, investigate later"** workflow for incident reporting.

What we've created:

- A new mental model: reporting doesn't have to be a 45-minute burden
- Handoff pattern: Report Agent → Investigation Agent → Expert Investigator
- Notification system: staff know when questions are waiting
- Dashboard-based follow-up: answer questions when convenient

Why it's a moat:

- Changing workflows is hard—once facilities adopt this pattern, switching back is painful
- The efficiency gains are immediately measurable (75% time reduction)
- Staff prefer this workflow, creating internal advocacy

3. Compliance-First Design

WAiK is built **around compliance requirements**, not retrofitted for them.

What we've done:

- Gold Standards define what "complete" looks like
- Every question maps to a compliance field
- Scoring is based on regulatory requirements
- Reports are structured for audit readiness

Why it's a moat:

- Competitors focused on features, not compliance
- Our system ensures facilities pass audits by design
- The compliance structure is deep in the architecture, not a surface feature

4. Continuous Learning Loop

WAiK is designed to **get better over time**.

Planned capabilities:

- Question quality improves from feedback
- Gold Standards refine based on real-world gaps
- Pattern analysis identifies systemic issues
- Predictive analytics prevent incidents before they occur

Why it's a moat:

- The system learns from every incident across all customers
- Network effects: more data → better AI → better outcomes → more customers
- Competitors start from scratch; we start from thousands of documented incidents

5. Healthcare-Specific Infrastructure

WAiK is built **specifically for senior care**, not a generic tool adapted to healthcare.

What we've built:

- Fall-specific incident types (wheelchair, bed, slip, lift)
- Senior care terminology and workflows
- Role-based access (staff, admin) matching facility structures
- Integration readiness for EHR systems (HL7/FHIR roadmap)

Why it's a moat:

- Generic tools require significant customization
- We speak the language of senior care
- Our roadmap includes healthcare-specific features (EHR integration, family portal)

Product Vision & Roadmap

Where We Are Now (V1 MVP)

✅ **Complete**:

- Two-agent architecture (Report Agent + Investigation Agent)
- Expert Investigator for conversational gap-filling
- Voice-enabled reporting (Companion mode)
- Gold Standards for fall incidents
- Compliance scoring and gap analysis
- AI-generated reports
- Intelligence Q&A (RAG)
- Staff and Admin dashboards
- Notification system

Where We're Going

Phase 2: Production Hardening (Q1 2025)

- MongoDB migration for scalability
- Rate limiting and error logging
- Session management and audit logging
- Enterprise security features

Phase 3: Advanced AI (Q2 2025)

- **Pattern Analysis**: Cross-incident insights
- **Risk Prediction**: ML-based scoring to identify at-risk residents
- **Voice Conversation Agent**: Replace script-based with true conversational AI
- **Trend Dashboard**: Analytics and visualization

Phase 4: Enterprise Features (Q3 2025)

- **Multi-Facility Support**: Tenant isolation for chains
- **SSO Integration**: SAML/OAuth for enterprise
- **EHR Integration**: HL7/FHIR for bidirectional sync
- **Mobile App**: Native iOS/Android experience

Phase 5: Compliance & Reporting (Q4 2025)

- **CMS Compliance Reports**: Regulatory export
- **Custom Report Builder**: User-defined reports
- **Scheduled Reports**: Email delivery
- **Bulk Export**: CSV/PDF for audits

Long-Term Vision (2026+)

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│    │    • Identify at-risk residents before incidents
│    │
│    │    • Suggest preventive interventions
│    │
│    │    • Facility-wide risk scoring
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│
│    │                    LEARNING SYSTEM
│    │
│    │    • Improve question quality from feedback
│    │
│    │    • Adapt to facility-specific patterns
│    │
│    │    • Continuous Gold Standard refinement
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│
│    │                    INTEGRATED ECOSYSTEM
│    │
│    │    • EHR bidirectional sync
│    │
│    │    • Family portal for updates
│    │
│    │    • Regulatory auto-reporting
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Aspect	Traditional Forms	WAiK
Input method	Typing/writing	Voice conversation
Time required	45-60 minutes	~10-15 minutes
Questions	Generic checklist	Context-specific, expert
Compliance	40-60% complete	90%+ complete
Workflow	Immediate, blocking	Asynchronous, non-blocking
Analysis	Manual review	AI-powered insights

vs. Generic Incident Management Software

Aspect	Generic Tools	WAiK
Domain knowledge	None	Deep senior care expertise
Fall classification	Manual	Automatic (wheelchair/bed/slip/lift)
Question generation	Static templates	AI-tailored to each incident
Compliance scoring	None	Real-time, weighted
Voice support	Limited/none	First-class citizen

vs. EHR-Integrated Incident Modules

Aspect	EHR Modules	WAiK
Focus	Generic healthcare	Senior care specific
UX priority	Documentation	Speed and simplicity
AI capabilities	Limited/none	Core to product
Agentic workflow	None	Two-agent architecture
Staff experience	Secondary	Primary design goal

Summary

WAiK is not just better incident reporting—it's a fundamental rethinking of how documentation should work in healthcare.

The Core Innovation

We separated what humans must do (observe and describe) from what AI can do better (analyze, classify, question, score). This simple insight enables:

- **75% reduction in staff time**
- **90%+ compliance completion**
- **Expert-level documentation from any staff member**

- **Real-time visibility for administrators**
- **Audit readiness by design**

The Moats

Technological:

1. Two-Agent Architecture
2. Gold Standards System
3. Intelligent Question Generation
4. Voice-First Experience
5. RAG-Powered Intelligence
6. Real-Time Compliance Scoring

Process-Based:

1. Expert Knowledge Embedded in Software
2. Asynchronous Workflow Pattern
3. Compliance-First Design
4. Continuous Learning Loop
5. Healthcare-Specific Infrastructure

The Vision

From reactive documentation to proactive prevention. WAiK's ultimate goal is not just to document incidents better, but to use the intelligence gathered to prevent them from happening in the first place.

WAiK: Wake up to better incident documentation.